

Main Checklist for Data_Project

Progress tracking for the DDR Data Logger project on Zybo Z7-20

Main Checklist

Stage	Stage Name	Status	Stage Output
0	GitHub + Project Structure	✓ Done	Repository, folders, README, docs, first commit
1	Data Protocol	✓ Done	docs/data_protocol.md with Header, Length, Data, Checksum
2	Data Generator + Testbench	✓ Done	rtl/data_generator.v, tb/tb_data_generator.v, simulation, waveform, report
3	Packet Builder FSM + Testbench	✓ Done	packet_builder_fsm.v + testbench + simulation
4	Sync FIFO + Testbench	✓ Done	sync_fifo.v + testbench + simulation
5	AXI-Stream Master + Testbench	□ Pending	axis_stream_master.v + testbench + simulation
6	Full RTL Top Simulation	□ Pending	data_project_top.v + full testbench + simulation
7	Vivado Block Design + AXI DMA + DDR	□ Pending	Connected Block Design ready to run
8	Vitis C Verification	□ Pending	C software that starts DMA and verifies DDR data
9	Async FIFO + CDC + Testbench	□ Pending	async_fifo.v + testbench with two clocks
10	Full Integration with CDC + DMA + DDR	□ Pending	Full system with two clock domains and DDR
11	ILA + Vivado Reports	□ Pending	ILA, timing report, utilization report
12	Python CSV / Graph	□ Pending	CSV, graph, and documentation
13	UART or SPI Extension	□ Pending	Real data source replacing Data Generator
14	Portfolio Polish	□ Pending	GitHub, README, images, explanations, and reports polish

Current status: Stages 0-4 are complete. The next stage is Stage 5 - AXI-Stream Master + Testbench.

Detailed Checklist for Work Tracking

Stage 0 - GitHub + Project Structure

✓ Done

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|--|--|
| ☒ Created Data_Project folder | ☒ Created work plan document |
| ☒ Created GitHub repository | ☒ Added block diagram |
| ☒ Created README.md | ☒ Completed first commit |
| ☒ Created organized folder structure | ☒ Completed push to GitHub |

Stage 1 - Data Protocol

✓ Done

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|---|--|
| ☒ Created docs/data_protocol.md | ☒ Defined Checksum |
| ☒ Defined DATA_WIDTH = 32 | ☒ Defined Expected Packet |
| ☒ Defined Header | ☒ Defined PASS / FAIL Criteria |
| ☒ Defined Length | ☒ Completed commit |
| ☒ Defined Data sequence | ☒ Completed push |

Stage 2 - Data Generator + Testbench

✓ Done

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|--|--|
| ☒ Defined block goal | ☒ Verified expected sequence: 1, 7, 8, 3, 6, 3 |
| ☒ Defined inputs and outputs | ☒ Saved waveform screenshot: images/
stage2_data_generator_waveform.png |
| ☒ Wrote rtl/data_generator.v | ☒ Created simulation report: reports/
stage2_data_generator_simulation.md |
| ☒ Wrote tb/tb_data_generator.v | ☒ Completed commit |
| ☒ Ran Behavioral Simulation | ☒ Completed push |

Stage 3 - Packet Builder FSM + Testbench

✓ Done

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|--|--|
| ☒ Define block goal | ☒ Run simulation |
| ☒ Define FSM states | ☒ Verify Header / Length / Data / Checksum |
| ☒ Write rtl/packet_builder_fsm.v | ☒ Complete commit |
| ☒ Write tb/tb_packet_builder_fsm.v | ☒ Complete push |

Stage 4 - Sync FIFO + Testbench

✓ Done

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|---|---|
| ☒ Define block goal | ☒ Verify full/empty |
| ☒ Write rtl/sync_fifo.v | ☒ Verify overflow/underflow |
| ☒ Write tb/tb_sync_fifo.v | ☒ Complete commit |
| ☒ Verify write/read | ☒ Complete push |

Detailed Checklist for Work Tracking

Stage 5 - AXI-Stream Master + Testbench

☐ Pending

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|---|--|
| ☐ Define block goal | ☐ Verify backpressure |
| ☐ Write rtl/axis_stream_master.v | ☐ Complete commit |
| ☐ Write tb/tb_axis_stream_master.v | ☐ Complete push |
| ☐ Verify tvalid / tready / tdata / tlast | |

Stage 6 - Full RTL Top Simulation

☐ Pending

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|---|--|
| ☐ Write rtl/data_project_top.v | ☐ Save result |
| ☐ Write tb/tb_data_project_top.v | ☐ Complete commit |
| ☐ Connect all blocks | ☐ Complete push |
| ☐ Run full simulation | |

Stage 7 - Vivado Block Design + AXI DMA + DDR

☐ Pending

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|--|---|
| ☐ Create Block Design | ☐ Connect DDR |
| ☐ Add Zynq PS | ☐ Pass design validate |
| ☐ Add AXI DMA | ☐ Complete commit |
| ☐ Add Interconnect / SmartConnect | ☐ Complete push |
| ☐ Connect Custom RTL | |

Stage 8 - Vitis C Verification

☐ Pending

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|--|---|
| ☐ Create Vitis project | ☐ Read data from DDR |
| ☐ Write DMA initialization code | ☐ Verify Header / Length / Data / Checksum |
| ☐ Define DDR buffer address | ☐ Complete commit |
| ☐ Start DMA transfer | ☐ Complete push |

Stage 9 - Async FIFO + CDC + Testbench

☐ Pending

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|---|---|
| ☐ Write rtl/async_fifo.v | ☐ Verify correct data crossing |
| ☐ Write tb/tb_async_fifo.v | ☐ Complete commit |
| ☐ Verify two clocks | ☐ Complete push |

Stage 10 - Full Integration with CDC + DMA + DDR

☐ Pending

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|---|---|
| ☐ Integrate Async FIFO into the system | ☐ Verify DDR write |
| ☐ Update full integration | ☐ Complete commit |
| ☐ Verify two clock domains | ☐ Complete push |

Detailed Checklist for Work Tracking

Stage 11 - ILA + Vivado Reports

☐ Pending

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|--|---|
| ☐ Add ILA | ☐ Save screenshots |
| ☐ Verify key signals | ☐ Complete commit |
| ☐ Save timing report | ☐ Complete push |
| ☐ Save utilization report | |

Stage 12 - Python CSV / Graph

☐ Pending

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|--|---|
| ☐ Create CSV file | ☐ Save screenshot / output |
| ☐ Write Python script | ☐ Complete commit |
| ☐ Create graph | ☐ Complete push |

Stage 13 - UART or SPI Extension

☐ Pending

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|--|---|
| ☐ Choose UART or SPI | ☐ Receive real data through the system |
| ☐ Write communication block | ☐ Complete commit |
| ☐ Run verification | ☐ Complete push |

Stage 14 - Portfolio Polish

☐ Pending

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|--|--|
| ☐ Update README | ☐ Add explanations for each block |
| ☐ Add images | ☐ Add Future Work / Lessons Learned |
| ☐ Add waveforms | ☐ Prepare project for presentation |
| ☐ Add reports | |